JOHNS HOPKINS
MECHANICAL ENGINEERING
SENIOR DESIGN

## Project Brief: JHUSPH27

# IMPROVED MOSQUITO MARKING AND TRAP DEVICE

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| Sponsor | Dr Derek Cummimgs, JHU School of Public Health |
| Contact | Derek Cummings, PhD<br>Bloomberg Distinguished Professor<br>Department of Epidemiology (JHSPH)<br>Department of Biomedical Engineering (WSE)<br>Johns Hopkins University<br>Email: derek.cummings@jhu.edu<br>Phone: 410-916-1371 |
| Team Citizenship | No Restriction |
| IP Ownership | Sponsor, student agreement required |
| Non-Disclosure Agreement | NDA Required |
| Budget | |

## Project Goal

The deadliest animal to humans in the world is the mosquito. Dr. Derek Cummings and his team at the School of Public Health are developing ways to combat the ability of mosquitoes to infect humans with deadly diseases. Part of that effort is to track populations and movements of mosquitoes in a given area through mark-release-recapture statistics; methods that have been used in ecology for years.

The goal of this project is to design and develop a field-deployable mosquito trapping system capable of attracting, speciating (with an optical AI system), marking (with fluorescent powder), and releasing mosquitoes in a controlled and repeatable manner. The system will return data to researchers to enable improved estimation of mosquito populations, tracking movement patterns, and data collection for disease surveillance.

## Background

Vector-borne diseases such as dengue, chikungunya, and Zika remain significant global health challenges. A critical limitation in controlling these diseases is the inability to accurately estimate absolute mosquito population sizes.

These applications are motivated by ongoing trial work of Dr. Cummings and his team. They are developing an intervention that introduces mosquitos infected with a bacteria (Wolbachia) that acts as a vaccine against dengue, chikungunya, and Zika. This trial is about to be unblinded in Belo Horizonte, Brazil. But in performing the trial and analyzing other deployments, his team found that an inability to estimate absolute numbers of mosquitos in environments (rather than an imperfect relative number) limits many interventions like this one. In addition, understanding of mosquito mobility and age structure are often critical to understanding the effect of specific interventions. Reductions in mosquito populations by intervention can be quickly erased by re-colonization of areas by surrounding populations. Interventions that shifted age structures of mosquitoes could have an outsized impact on pathogen transmission as many pathogens need to go through a life cycle in the mosquito, requiring days to weeks. Interventions that reduced older mosquitoes could target those that are more likely to transmit a pathogen. These details of the mosquito-pathogen-human interaction are difficult to measure.

Dr. Cummings lab has developed some early models of the catch and release system which may help the team at the start of the project. The current device incorporates elements such as 3D printing, optics, embedded systems (e.g., Raspberry Pi), and AI-based analysis.

## Opportunities

- Work with interdisciplinary teams spanning engineering and public health
- Access to lab and prototype facilities within JHU
- Opportunity to contribute to global health research with real-world deployment potential

## Requirements

1. The system shall attract mosquitoes effectively using appropriate stimuli. Effectiveness will be measured as the percent of reared and released mosquitos captured in an enclosed environment. Maintaining or improving capture rate of current rate is required.
2. The system shall apply consistent fluorescent marking without harming mosquitoes. Marking will be measured by a validated index of measured body parts with detectable powder. Injury rate will be the percentage of mosquitos attracted to trap that experience injury. Maintaining injury rate and improvement of marking rate is required.
3. The system shall release mosquitoes safely following marking.
4. The system shall count mosquitoes with an accuracy of 90%
5. The system shall speciate mosquitoes with an accuracy of 70%
6. The system shall collect and store acquired data with an easy method for data extraction.
    - Stretch goal: system shall transmit collected data via Wi-Fi.
7. The system shall have a deployment duration (time between human intervention?) of 7 days.
8. The device shall be portable by one person for field deployment and be contained on a platform to be placed on the ground.
9. The system shall operate with an onboard power supply (replacing existing 12V DC and 6V DC power source).
    - Stretch goal: system shall use a rechargeable solar power source
10. The system shall be robust to environmental conditions:
    - Operate in rain
    - Operate in temperature range of 40 to 120F and in direct sunlight in equatorial locations
    - Be debris resistant over the intended deployment duration

## Project Milestones

1. By Week 4 present a trade study of possible system approaches to the sponsor. Decide on one or more paths to pursue further. To accomplish this, you must:
    a. Become mosquito experts.  Understand what attracts them, how they fly (poorly!), how to interact with them without damaging them, etc.
    b. Meet with the research team and investigate the existing solutions and their features, functions, strengths, and weaknesses.
    c. Conduct a design review of the existing trap design
    d. Some members of the team should become experts in methods to identify the mosquito species using AI. What models exist? Identify the gaps.
    e. Quickly develop and prototype ideas.
2. By Week 6 conduct a System Requirements Review (SRR) with the sponsor, capturing all system level requirements, including performance requirements that were underdefined above.

3. By Fall Design Day, develop a proof-of-concept prototype. This prototype may not address all requirements but must present a low-risk path toward modifying the design to meet the full range of requirements. You must also have an approach for testing, including any special test equipment you need to buy / build.
4. By Spring Break conduct a TRR – Test Readiness Review - that will outline the test plans and test equipment you will use to verify your design meets the requirements. TRR should contain a schedule you intend to follow, facilities needed, and the timeline of when all test equipment will be ready for use.
5. By April 1 the assembly of the final hardware must be complete, and final testing must begin for the design requirements.
6. Most of Spring should be about perfecting designs based on feedback and testing data. Execute your test plans and verify performance against the technical requirements.
7. By Spring Design Day complete all testing, including customer-witnessed (or customer conducted) testing at their facility if possible. All hardware changes based on testing must be complete and incorporated.
8. Before project closeout, prepare and submit all documentation and hardware in the Deliverables section below.

## Technology/Skills

- Mechanical design for small-scale biological systems
- Fluid/particle handling (e.g., powders, airflow, mosquito flight mechanics)
- 3D printing and rapid prototyping
- Embedded systems (Raspberry Pi, sensors)
- Optical detection and imaging systems coupled with AI processing
- Experimental design and statistical analysis
- Interdisciplinary collaboration (engineering + public health)
- Field-deployable product design

## Deliverables to the Sponsor

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| CAD Models | Full assembly of your device in Solidworks, delivered in STEP or other format if requested. |
| Drawings and Schematics | Critical dimensions and tolerances of mechanical parts.<br><br>Assembly drawings, (may use exploded views) etc that show how assemblies of piece parts and/or subassemblies go together. Include special assembly instructions as notes on the drawing.<br><br>Electrical schematics, PCB drawing etc |

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| Bill of Material (BOM) | Complete parts list of your design, in Excel, with unit and total cost.<br>Include all parts in your design, so that it can be repeatably built and all parts purchased to your documentation. |
| Source Code | if applicable |
| Operation Manual | Step by step instructions on how to use your product. Write it for a typical intended user, not an engineer (ie your team or the sponsor design lead). |
| Maintenance Manual | Provide instruction so that the typical user can keep your product alive and running well. |
| Physical prototype(s) | The system(s) you built for the project, with the sponsors funding. May include all iterations. May include a set of spare parts. |
| Test Report(s) | Prepare a report presenting device performance results, baselined, if possible, to competing solutions. |